

Three-Layer Architecture: Ontology, Dynamics, Execution (Commit Gating)

Alexander Wolfgang Strüver

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Thailand

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1 Purpose

This document defines a strict three-layer separation:

- 1) **Ontology Layer**: what exists (domain-invariant primitives).
- 2) **Dynamics Layer**: how states evolve (domain equations and projections).
- 3) **Execution Layer (Commit Gating)**: what is allowed to execute at irreversible boundaries.

The separation is a methodological constraint: **no layer is allowed to smuggle semantics from another layer.**

2 Layer Overview (Definitions)

2.1 Layer 1: Ontology Layer (Primitive Commitments)

Goal: define what is ontologically prior and domain-invariant.

Ontological primitives (example set).

- **Energy** as transition-capacity (transition possibility).
- **Mass** as energy-density (bound transition-capacity per extent).
- **Space** as relation among energy-density distributions (not a container).
- **Process/Ordering** (“time”) as operator semantics (e.g., Φ) rather than a dimension.

Output of this layer.

- Axioms/definitions only.
- Permitted inference types: definitional consequences, consistency constraints, admissible forms of projections.
- **No** domain proxy selection, **no** numerical calibration, **no** commit semantics.

2.2 Layer 2: Dynamics Layer (Equations and Observable Projections)

Goal: connect ontological primitives to testable consequences under frozen protocols.

What lives here.

- Domain equations, approximations, and perturbative corrections derived from the ontological commitments.
- Observable projections: measurable relations (e.g., ordering constraints, correction terms, scaling laws).
- Rejection conditions: if projection fails under pre-registered pipelines, reject the projection (or domain mapping).

Output of this layer.

- Predicted behaviors under specified assumptions.
- Falsification protocols (frozen extraction, fixed constants, no per-event tuning).
- Still **no** execution permission logic; it only predicts/relates observables.

2.3 Layer 3: Execution Layer (USK / UPASL Commit Gating)

Goal: enforce safety and admissibility at **irreversible commit boundaries**.

Core commitments.

- **Observable state** $x(t) \in \mathbb{R}^n$ exists as *measured* telemetry (not ontology).
- **Evidence validity:** measurements are admissible only if freshness and conformance constraints hold.
- **Irreversible commit:** some actions cross a boundary beyond which the trajectory cannot be deterministically reverted within the operational horizon.

Output of this layer.

- Deterministic decisions at commit time (e.g., ALLOW/LIMIT/REFUSE).
- Fail-closed behavior: missing evidence implies maximal risk / refusal.
- Non-bypassable gating rules bound to invariants and hazard.

3 Explicit Dependency Rules (Non-Negotiable)

3.1 Allowed dependencies

- **Dynamics** may depend on **Ontology** (primitives constrain admissible equation forms).
- **Execution** may depend on **Dynamics** only via **frozen, public-facing projections** (e.g., thresholds, invariant envelopes, ordering constraints), never via internal physics speculation.
- **Execution** may depend on **Ontology** only indirectly (e.g., justification of “why determinism/falsifiability matters”), not as a substitute for measurable evidence.

3.2 Forbidden dependencies

- **Ontology** must not depend on **Dynamics** outcomes (no retrofitting axioms to match data).
- **Ontology** must not depend on **Execution** semantics (commit logic is not ontology).
- **Dynamics** must not import **Execution** decisions as physical claims (“REFUSE” is not a physics prediction).
- **Execution** must not pretend to be **subsystem physics** (it is an admissibility gate).

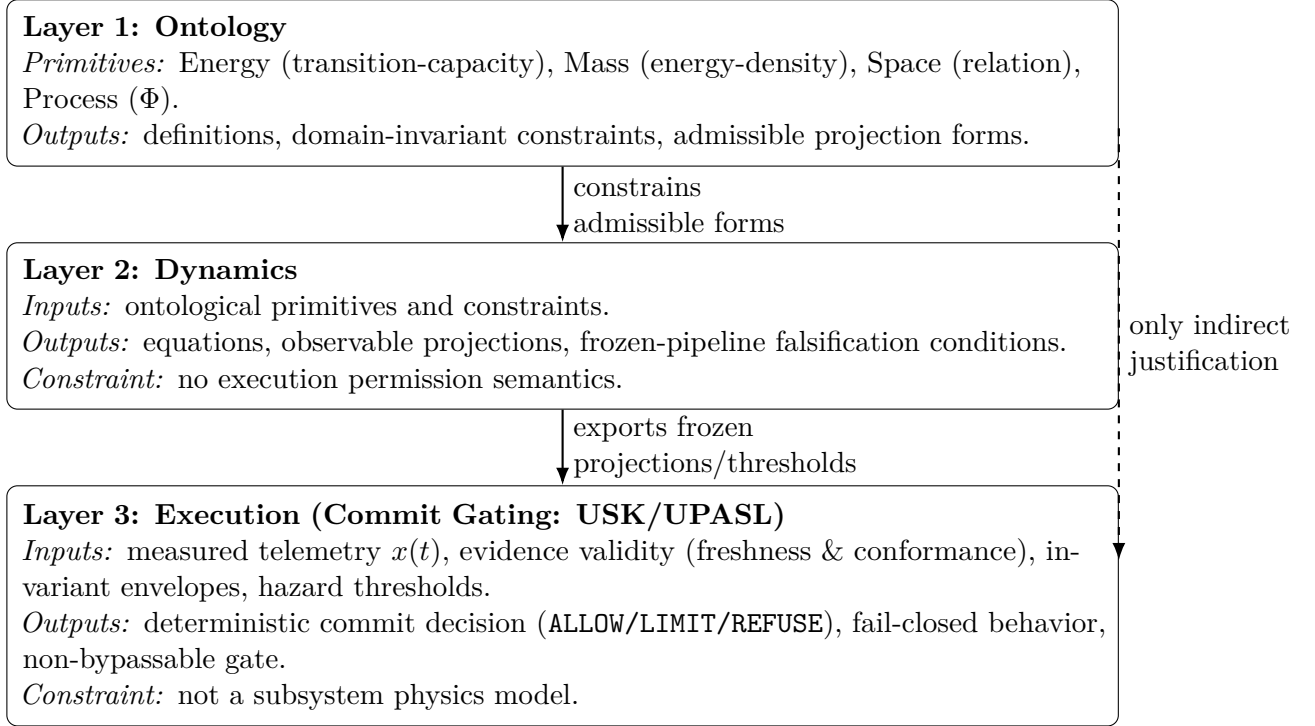
4 Relational Interpretation (Why Different “Axioms” Appear)

The word “axiom” is used in two senses:

- a) **Ontological axioms:** claims about what exists and what is primary.
- b) **Admissibility axioms:** constraints required for safe execution under limited evidence.

These sets are not in conflict because they live in different layers with different truth-conditions: ontology is evaluated by conceptual coherence and falsifiable projections; execution is evaluated by safety, determinism, and non-bypassability at irreversible boundaries.

5 Architecture Diagram (Dependencies and Interfaces)



6 Minimal Interface Contract (One Paragraph)

Interface contract: The Execution Layer consumes only (i) validated evidence atoms, (ii) invariant envelopes and limits, and (iii) frozen projections/thresholds derived from the Dynamics Layer. It does not consume ontology as “physics” and does not modify domain laws. The Ontology Layer never changes based on execution outcomes. Any violation of these constraints is a category error and invalidates falsifiability.

7 Checklist for Consistency Audits

1. Are ontology primitives stated without reference to specific datasets or events?
2. Are dynamic projections testable under a frozen protocol with explicit rejection criteria?
3. Does commit gating refuse when evidence is missing (fail-closed) and remain non-bypassable?
4. Is there any place where commit decisions are described as physical claims? (must be **no**)
5. Is there any place where physics axioms are changed to fit an observed event? (must be **no**)